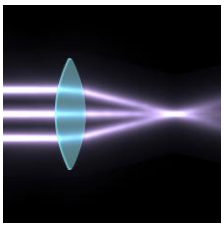
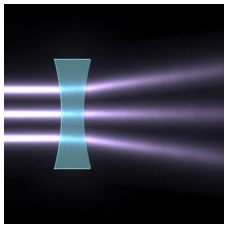
Chapter 60 Imaging using Lenses
Calculus Physics II

Refraction is the bending of light due to the change in speed of the light going from a material with a different index of refraction than the material the light is going into. This bending of light can be used to produce a focused image of an object.







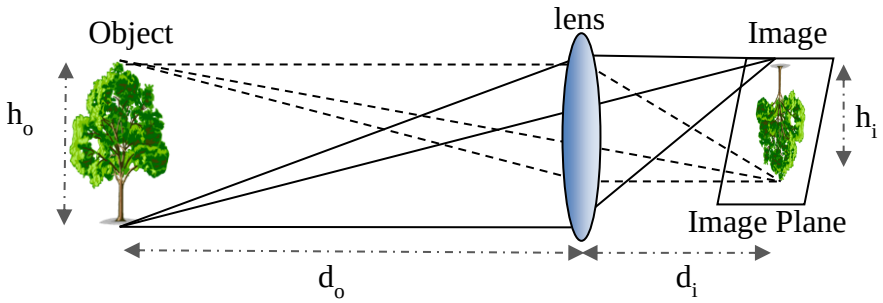
Lenses – The index of refraction of the material bends the light in order to either converge the light rays or diverge the light rays to form the image of the object.

Object – is the source of the light rays by either the object emitting light itself (light bulb) or the light being reflected off of the object and is what is being imaged.

Image – The appearance of a reproduction of the object that your eye sees, it is formed when the light rays are reflected by a mirror or refracted by a lens. It is technically defined as the collection of focus points where light rays originating from an object either actually meet or appear to meet after passing through an optical system.

Real Image – An image from which rays of light actually emanate, the image can be seen on a screen if placed at the image plane.

Light from the tree (object) are focused on to the piece of paper to form a real image of the tree. Light from all points on the tree travels through the lens and forms the image of each point of the tree on the paper, so each point on the tree (object) emits a bundle of light rays that are focused back to a point on the paper (image).



Object Distance (s_o or d_o) – The distance from the mirror to the object that is being imaged.

Image Distance (s_i or d_i) – the distance from the mirror to where the image appears in focus.

Object height (h_o) – the actual height of the object.

Image Height (h_i) – the height that the image appears to have.

Image Plane – Imaginary plane in which the image appears in focus.

Linear Magnification (m) – the ratio of the image height to the object height and indicates the change in sizes of the image relative to the object.

Image formation for lenses: For lens the real image is formed on the opposite side of the lens as object because the light rays are transmitted through the lens and then converge to form the image on the opposite side of the lens than the object. Virtual images are formed on the same side of the lens as the object. This is the major difference between mirror and lens image formation.

Virtual Image – An image from which rays of light do not actually come, but only appear to do so, no image will appear on screen if placed at the image plane. The image is not where it appears to be. Your eye can not tell the difference between a real or virtual image.

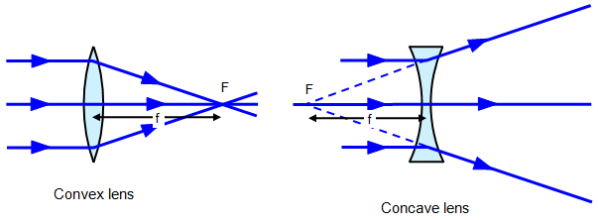
The stamp is placed inside the focal length of the lens so that light from the stamp goes through the lens to the eye of the person. If we place a piece of paper on the other side of the lens, we will not see the image of the stamp appear on the paper.



- For lenses virtual images appear on the same side as the object.
- Must look into the optical system to see a virtual image.

Principle Focal Point – of a thin lens with a spherical surface is the point where rays parallel to and near the principle axis are brought to a focus or appear to focus: this focus is real for a converging lens and virtual for a diverging lens.

- ❖ It is also where an image will be formed for an object that is very far (infinite distance) from the convergent lens.
- ❖ For a convergent lens, the principle focal point is on the opposite side of the lens as the object and the direction of the light rays.
- ❖ For a divergent lens the principle focal point is on the same side of the lens as the object and direction of the light rays and is virtual.
- ❖ Because for a thin lens the direction of the light can be reversed without altering the rays, two symmetric focal points exist for each lens.

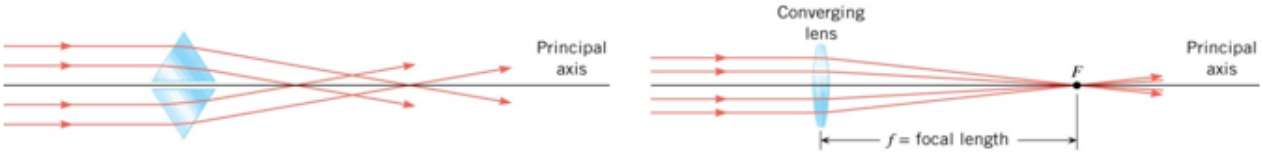


Secondary Focal Point – is the focal point that is located on the opposite side of the lens as the principle focal point. The Secondary focal point is the same distance from the lens as the principle focal point.

Focal length (f) – is the distance between the focal point and the center of the lens measured along the principle axis.

Converging Lens (Positive or Convex) – Converging lenses are thicker at the center than at the rim and will converge a beam of parallel light to a single point (the focal point).

Lens use the properties of refraction to bend light, and two prisms can act as a crude lens.

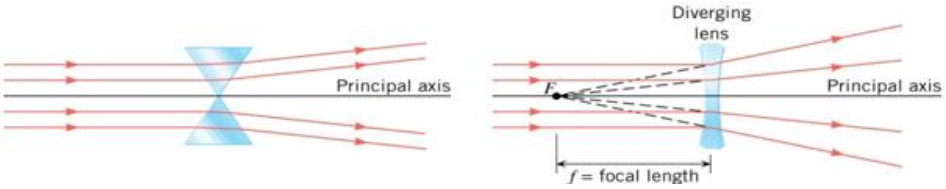


For a Converging lens and the image, they create.

- They can create real images that are inverted and either smaller or larger depending on where the object is placed outside the focal point (not between the lens and the focal point).
- They can create virtual images when the object is placed between the lens and the focal point, this virtual image is upright and larger than the object.

Diverging Lens (Negative or Concave) – Lenses are thinner near the center than at the rim and will diverge a beam of parallel light from a virtual focus.

Lens use the properties of Refraction to bend light, and two prisms can act as a crude lens.



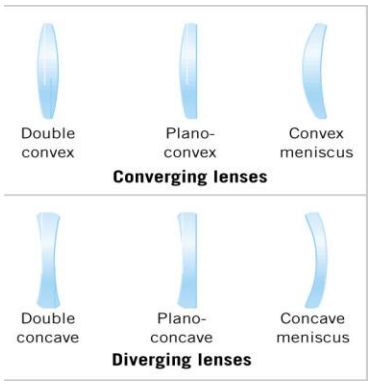
- For a diverging lens, the image formed is <u>always</u> virtual, upright and smaller than the object.

There are many types of convergent and diverged lenses depending on the application.

<u>Double lens</u> – curve on both sides and are used for short focal lengths where a lot of bending is required.

<u>Plano lens</u> – have just one curved surface and are the most common and the least expensive.

<u>Meniscus lens</u> – are the highest quality and the most expensive but create very high-quality images, most lenses in cameras use meniscus lenses.

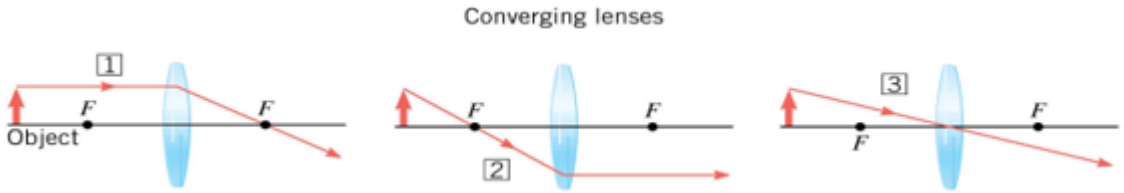


<u>Ray Tracing for Converging Lens</u>: - Using three light rays from a single point at the top of the object.

I. Light rays parallel to the principal axis and incident on a convergent lens will go through the focal point on the opposite side of the lens as the object.
II. Light rays going through the focal point that is on the same side as the object will come out of the lens parallel to the optical axis on the other side.
III. Light going through the center of the lens at the principal axis will not be bent.

(a) Ray from top of object parallel to principal axis passes through lens and is focused through focal point of lens.
(b) Ray from top of object passes through focal point of lens (focal point on same side of lens as object) passes through lens and is bent parallel to principal axis.
(c) Ray from top of object passes through center of lens at principal axis and keep going straight.

The Image of the point on the object is the intersection of these three lines.



<u>Examples of Ray Tracing using Convert Lenses</u>

1) Object distance (d_o) is more than twice the focal length (2f) from lens.

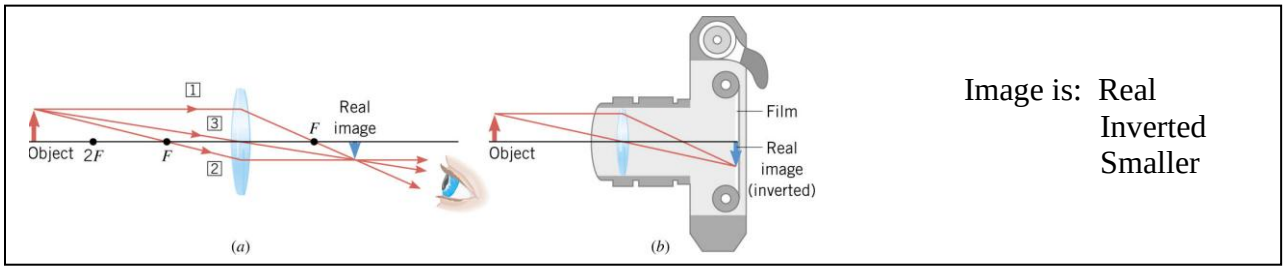


Image is: Real
Inverted
Smaller

2) Object distance is between the point that is twice the focal length (2f) from the lens and the focal length of the lens (f).

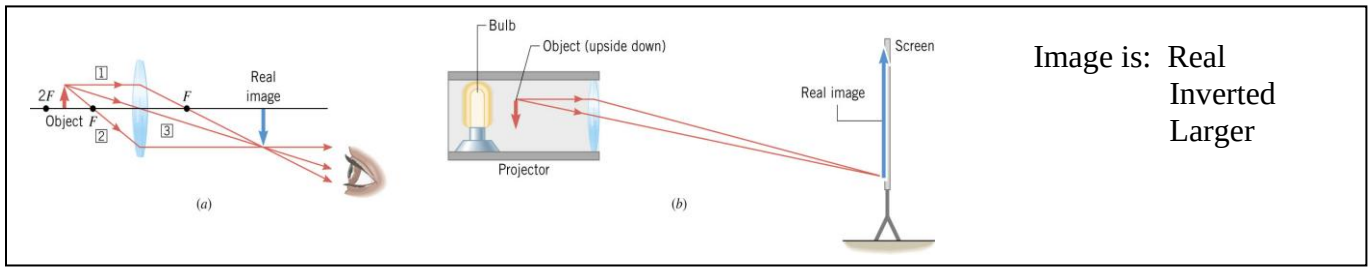


Image is: Real
Inverted
Larger

3) The object distance is between the focal length of the lens and the lens.

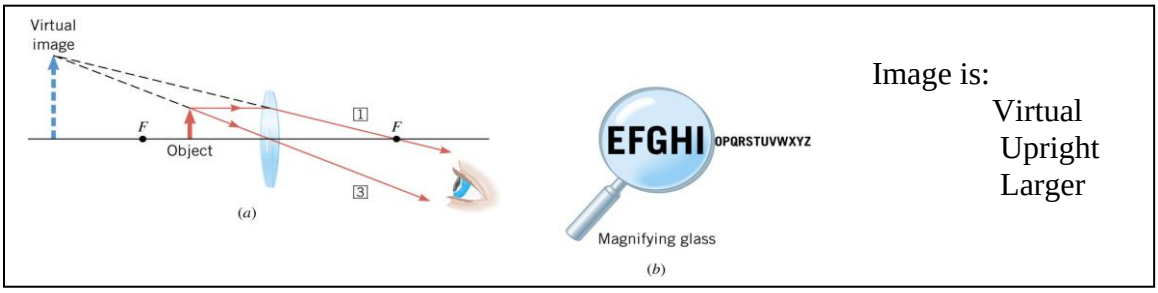


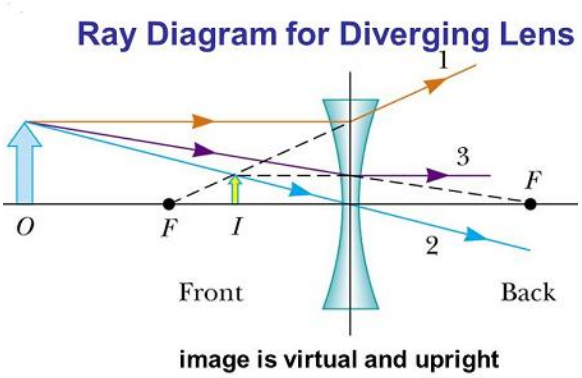
Image is:
Virtual
Upright
Larger

<u>Ray Tracing for Diverging Lens</u> - Using three light rays from a single point at the top of the object.

I. Light rays parallel to the principal axis and incident on a divergent lens will diverge on the opposite side of the lens but appear to originate from the focal point on the same side of the lens as the object.
II. Light rays going through the focal point that is on the opposite side as the object will come out of the lens parallel to the optical axis on the opposite side as the object.
III. Light going through the center of the lens at the principal axis will not be bent

1) Ray from the top of object parallel to principal axis passes through lens and is bent away from the principal axis as though it originated from the focal point of lens on the same side of the lens as the object.
2) Ray from the top of the object travels towards the focal point on the opposite side of the lens as the object but when reaching the lens, it travels parallel to the principle axis.
3) Ray from the top of object passes through center of lens at principal axis and keep going straight and are not bent.

The <u>refracted rays</u> are traced back and the Image of the point on the object is the intersection of these lines.



Example: For a divergent lens the object will always produce a virtual upright and smaller image.

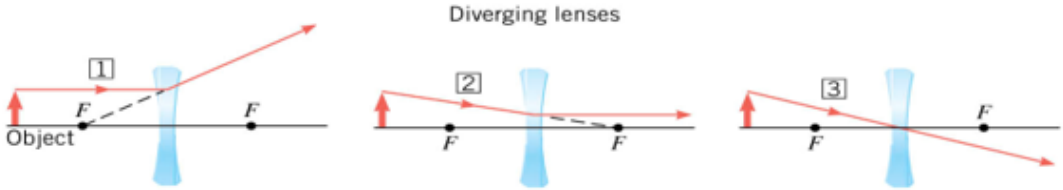
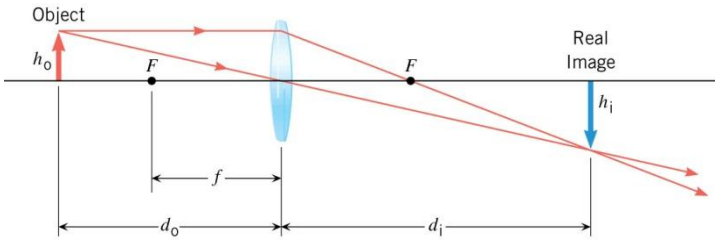


Image is:
Virtual
Upright
Smaller

Thin Lens equation – The lens equation relates the three main variables of the lens and image; focal length (f), object distance (d_o) and image distance (d_i). Given two the third can be calculated.

$$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$$



d_o – object distance*
d_i - image distance*
f – focal length* of lens

* All lengths are measured from the center of the lens along principle axis.

Calculating the image distance given focal length and object distance

$$d_i = \frac{1}{\left[\frac{1}{f} - \frac{1}{d_o}\right]}$$

d_o – is positive for a real object and negative for a virtual object.
d_i - is positive for a real image and negative for a virtual image.
f - is positive for a convergent lens and negative for a divergent lens.

Linear Magnification (m) of thin lens – Measures the change in size of the image relative to the object. The ratio of the image height to the object height.

$$m = \frac{h_i}{h_o} \quad \text{therefore the height of the image is given by } h_i = mh_o$$

Using the object and image distances the magnification is given as

$$m = -\frac{d_i}{d_o}$$

m is "+" for an image that is upright with respect to the object.
m is "-" for an image that is inverted with respect to the object.

Note: If the image is virtual then the image distance is negative when using the lens equation for calculations. If the image is real, then the image distance is positive.

- Magnification depends on the focal length of the lens
- Magnification depends on where the objects is placed relative to the focal point.

For a single lens system (and only a single lens)

- ✓ If the magnification is positive, the image distance must be negative, this mean the image is virtual and formed on the same side as the object.
- ✓ If the magnification is negative, the image distance must be positive, this means the image is real and formed on the opposite side as the object.

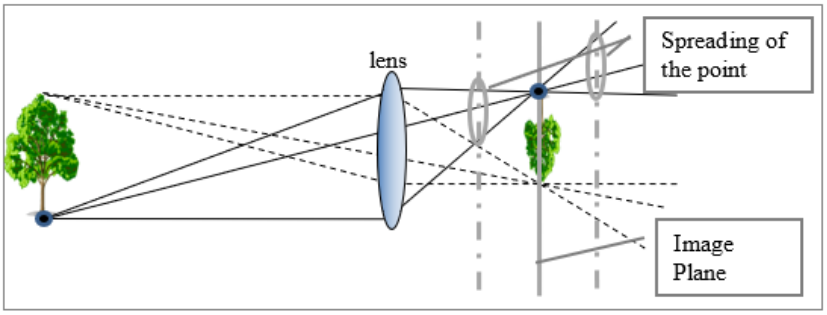
m > 1 image larger than object
m = 1 image same size as object
m < 1 image is smaller than object

m is negative then image inverted
m is positive then image is upright

Image Plane – When light from an object is incident on a convergent lens that light is focused on the opposite side of the lens as the object as long as the object is outside the focal point of the lens. Each point on the object is focused at the image plane, therefore if a piece of paper is placed at the image plane the image is clear and in focus. If the paper is moved towards the lens or away from the lens, then the image becomes blurry and out of focus.

- ✓ The image is only in focus at the image plane which is a two-dimensional plane parallel to the lens (perpendicular to the principle axis) and intersecting the principle axis at a single point.

The figure below shows only two points on the tree (object) that is the source of the light that travels through the lens and forms the image of the tree, there are billions and billions of more points but only two are shown for clarity. For each point on the tree there is shown only three rays of light but there are billions and billions of rays of light that fill the space between the top and bottom ray. The light from the single point on the tree is focused at a single point on the image plane. It can be seen as the paper is moved the rays form that single point, and the image is in focus. If the sheet of paper is moved towards or away from the lens, then the light rays emanating from a single point of on the object (the tree) do not form a single point on the paper but are spread out over an area causing the image to appear blurry.



Summary of Sign Conventions for Lenses

Focal Length

f is "+" for a converging lens
f is "-" for a diverging lens

Object distance (d_o)

d_o is "+" if the object is in front of the lens (real object).
d_o is "-"if the object is behind the lens (virtual object), this only happens when there is more than one lens.

Image distance (d_i)

d_i is "+" if the image is formed on the opposite side of the lens as the object (real image).
d_i is "-" if the image is formed on the same side of the lens as the object (virtual image).

Magnification

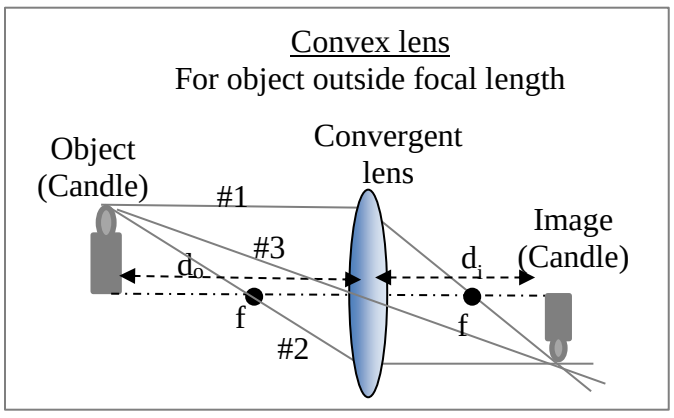
m is "+" for an image that is upright with respect to the object
m is "-" for an image that is inverted with respect to the object

Examples: For an object (h_o = 5 cm) placed 100 cm in front of a convergent lens with a focal length of 10 cm, what are the properties of the imaged formed?

1) Calculate image distance:

$$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$$

$$d_i = \frac{1}{\left[\frac{1}{f} - \frac{1}{d_o}\right]} = \frac{1}{\left[\frac{1}{10\ \text{cm}} - \frac{1}{100\ \text{cm}}\right]} = +11.1\ cm$$



✓ A positive image distance means that the image is real and on the opposite side of the lens as the object.

2) Calculate magnification:

$$m = -\frac{d_i}{d_o} = -\frac{11.1\ cm}{100\ cm} = -0.11$$

- ✓ The negative on the magnification indicates the image is inverted relative to the object, the image is upside down.
- ✓ the magnification is less than 1 therefore the image is smaller than the object by a factor of 0.11

3) Calculate the image height:

$$h_i = mh_o = (0.11)(5\ cm) = 0.55\ cm$$

- In summary the image is real, inverted, smaller (h_i = 0.55) than the object and formed 11.1 cm from the lens and on the opposite side of the lens as the object.

Example: For an object (h_o = 5 cm) placed 12 cm in front of a convergent lens with a focal length of 10 cm, what are the properties of the imaged formed?

1) Calculate image distance:

$$d_i = \frac{1}{\left[\frac{1}{f} - \frac{1}{d_o}\right]} = \frac{1}{\left[\frac{1}{10\ cm} - \frac{1}{12\ cm}\right]} = +60\ cm$$

- ✓ A positive image distance means that the object is real

2) Calculate magnification:

$$m = -\frac{d_i}{d_o} = -\frac{60\ cm}{12\ cm} = -5$$

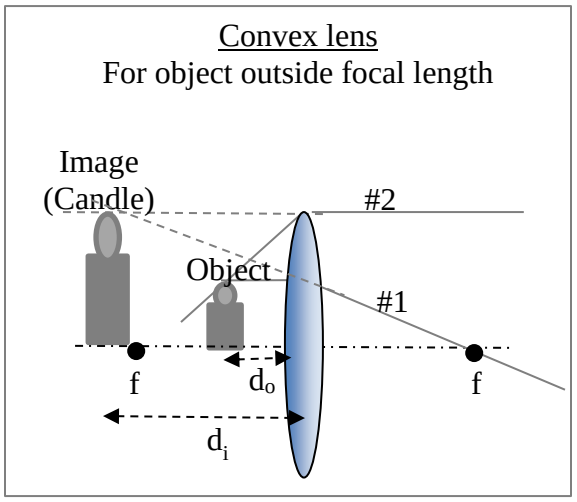
- ✓ The negative on the magnification indicates the image is inverted relative to the object, the image is upside down.
- ✓ the magnification is greater than 1 therefore the image is larger than the object by a factor of 5

3) Calculate the image height:

$$h_i = mh_o = (5)(5\ cm) = 25\ cm$$

- In summary the image is real, inverted, larger (h_i = 25 cm) than the object and formed 60 cm from the lens on the opposite side of the lens as the object.

Example: For an object (h_o = 5 cm) placed 6 cm in front of a convergent lens with a focal length of 10 cm, what are the properties of the imaged formed?



1) Calculate image distance:

$$d_i = \frac{1}{\left[\frac{1}{f} - \frac{1}{d_o}\right]} = \frac{1}{\left[\frac{1}{10\text{ cm}} - \frac{1}{6\text{ cm}}\right]} = -15\text{ cm}$$

- ✓ A negative image distance means that the image is virtual and appears to be on the same side of the lens as the object.

2) Calculate magnification:

$$m = -\frac{d_i}{d_o} = -\frac{-15\ cm}{6\ cm} = +2.5$$

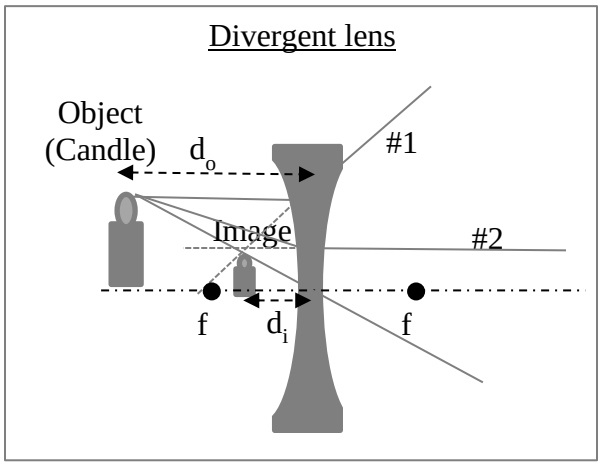
- ✓ The positive on the magnification indicates the image is upright relative to the object.
- ✓ the magnification is greater than 1 therefore the image is larger than the object by a factor of 2.5

3) Calculate the image height:

$$h_i = mh_o = (2.5)(5\ cm) = 12.5\ cm$$

- In summary the image is virtual, upright, larger than the object (h_i = 12.5 cm) and appeared to be formed 15 cm on the same side of the lens as the object.

Example: For an object (h_o = 5 cm) placed 5 cm in front of a divergent lens with a focal length of negative 20 cm, what are the properties of the imaged formed?



1) Calculate image distance:

$$d_i = \frac{1}{\left[\frac{1}{f} - \frac{1}{d_o}\right]} = \frac{1}{\left[\frac{1}{-20\text{ cm}} - \frac{1}{5\text{ cm}}\right]} = -4\ cm$$

- ✓ A negative image distance means that the image is virtual

2) Calculate magnification:

$$m = -\frac{d_i}{d_o} = -\frac{-4\ cm}{5\ cm} = +0.8$$

- ✓ The positive magnification indicates the image is upright relative to the object.
- ✓ the magnification is less than 1 therefore the image is smaller than the object by a factor of 0.8

3) Calculate the image height:

$$h_i = mh_o = (0.8)(5\ cm) = 4\ cm$$

- In summary the image is virtual, upright, smaller than the object (h_i = 4 cm) and appeared to be formed 15 cm on the same side of the lens as the object.

Example: If a candle is placed 40 cm from a lens and an image of the candle is formed 13.33 cm on the opposite side of the lens than the object, what type and what is the focal length of the lens?

$$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$$

$$f = \frac{1}{\left[\frac{1}{d_i} + \frac{1}{d_o}\right]} = \frac{1}{\left[\frac{1}{13.33\ \text{cm}} + \frac{1}{40\ \text{cm}}\right]} = +10\ cm\ \ and\ convergent\ lens$$

Example: If a candle is placed 30 cm from a lens and an image of the candle is formed 10 cm on the same side of the lens as the object, what type and what is the focal length of the lens? Because d_i is on the same side of the lens as the object it must be negative!

$$f = \frac{1}{\left[\frac{1}{d_i} + \frac{1}{d_o}\right]} = \frac{1}{\left[\frac{1}{-10\ \text{cm}} + \frac{1}{30\ \text{cm}}\right]} = -\ 15\ cm\ \ and\ divergent\ lens$$

Example: If an object is placed 40 cm in front of a single lens and the image has a magnification of m = + 0.4 what is the image distance, focal length and type of lens?

$$m = -\frac{d_i}{d_o}\ \text{therefore}\quad d_i = -md_o = -(+0.4)(40\ cm) = -16\ cm$$

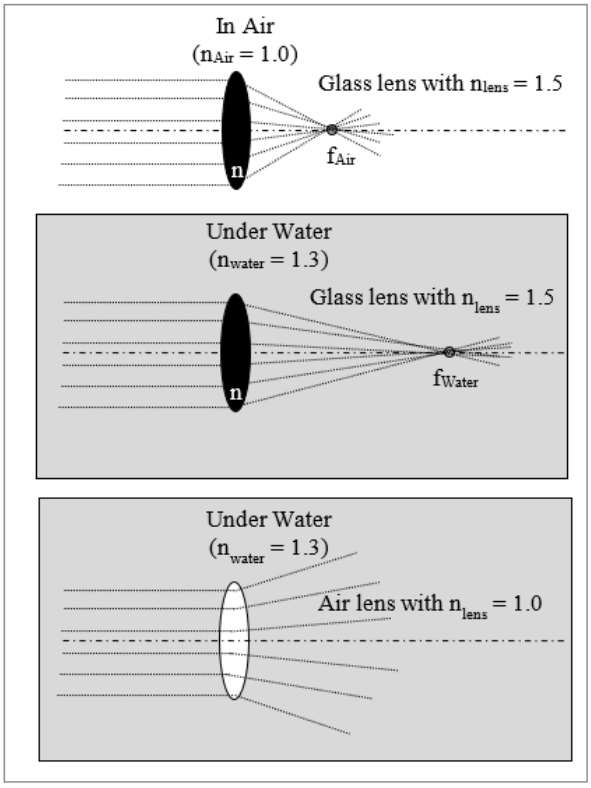
- For an image distance that is negative the image is virtual and on the same side of the lens as the object.

$$f = \frac{1}{\left[\frac{1}{d_i} + \frac{1}{d_o}\right]} = \frac{1}{\left[\frac{1}{-16\ \text{cm}} + \frac{1}{40\ \text{cm}}\right]} = -26.7\ cm$$

- ✓ A negative focal length indicates that the lens is a divergent lens.

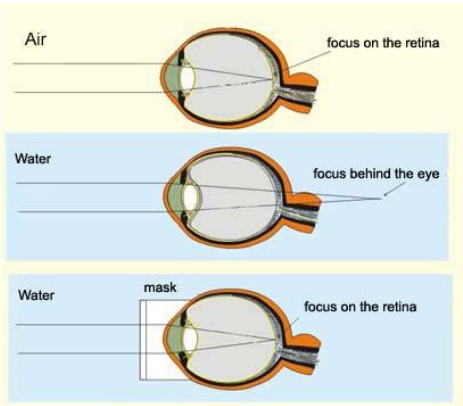
Lens in a liquid – A glass lens that placed in a substance other than air, the change in index of refraction at the interface will not be as large therefore there will not be as much bending of the light. The focal length will be changed; the focal length will be longer in the liquid than in air because it does not refract/bend the light as much as in air.

- For the glass convergent lens in air the air glass interface has an index change from 1.0 to 1.5 and the lens converges the light at the focal point.
- For the glass lens in water the index change is now l.33 to 1.5 there is not as large of change in index, the light is converged but not as much, this results in a longer focal length.
- If the glass lens is replaced with a hollow lens filled with air, then at the interface of the water lens is from 1.33 (water at high index) to air 1.0 (air at lower index), the light rays will be diverged, this acts as a divergent lens.
- If the lens had the same index of refraction as the water, then there would be no bending of the light rays and the light would go through the lens as if there was nothing there.



Example: When you open your eyes under water everything is burry, the water changed the amount the light bends entering your eye so that the image is no longer at the retina but has been moved farther from the lens of your eye. You would need very strong convergent lens to see under water.

When the eye is in air the refractive index change is large (n_{air} = 1 to n_{eye} = 1.4) but in water the change is not as large and the lens does not have the same refraction power (n_{water} = 1.33 and n_{eye} = 1.4) this is why it is diffiecut to see under water with out a mask. This also goes for fish, dophins or whales, there eyes are adaped for water eye interface so when they are out of the water they can not see as well.



Refractive Power (P) – The measure of the focusing strength of the lens, where f is the focal length in meters, this is what is used to specify eyeglasses.

P = 1/f Units are Diopters (D)

- ► Refractive power of a converging lens is positive and for a diverging lens is negative.
- ► The shorter the focal length the larger the refractive power.

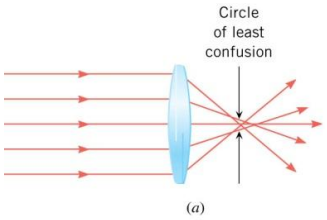
Example: Your eye doctor tells you that your glasses have a power of - 0.25 D, what type of lens and what is its focal length?

P = 1/f therefore f = 1/P = 1/(-0.25) = - 4 m

- A negative focal length means it is a divergent lens.

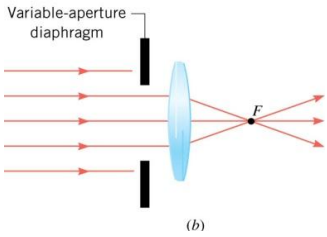
Lens Aberrations (Deviation from true image) – The quality of the image formed by a lens or lens system depends not only if the image is formed on the image plane but also the curvature of the lens and the frequency of the light. The aberration of the image can be caused by either the curvature of the lens or dispersion of the light through the lens.

Spherical Aberration – An effect in which rays from the outer edge of a spherical lens are not focused at the same point as rays that passing close to the center of the lens, thus causing images to be blurred.



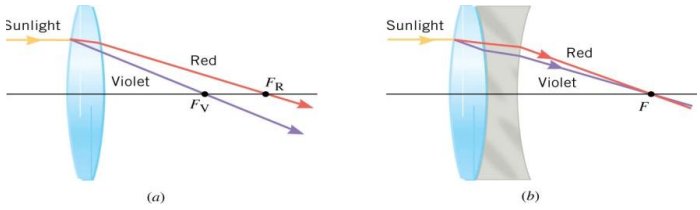
Circle of least confusion – is the position that gives the highest quality image.

Aperture – device used to minimize spherical aberration by limiting the outer rays that get to the lens, allowing only the paraxial rays to pass.



Chromatic Aberration – An undesirable lens effect arising when a lens focuses different colors of light at different points due to the different index of refraction for frequency of light.

Compound Lens – two lenses, one is the convergent lens and the second is a divergent lens which brings the colors back to the focus. Minimizes chromatic aberrations.



- ➢ A Meniscus lens can reduce both Spherical and Chromatic Aberrations.